

# **Intelligent Decision Support System for Sri Lankan Tea Estate Management**

R26-IT-118

Project Proposal Report

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B.Sc. (Hons) Degree in Information Technology

Department of Information Technology

**Sri Lanka Institute of Information Technology**

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Supervised By: Mrs. Uthpala Samarakoon

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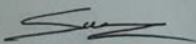
## Declaration

### DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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## Abstract

Effective labor management in Sri Lankan tea estates is a complex challenge that directly impacts both worker welfare and estate productivity. Traditional labor administration in tea plantations relies heavily on manual methods for attendance tracking, task assignment, and grievance handling. These manual processes are prone to inefficiency, unequal workload distribution, and delays in resolving worker concerns. Furthermore, there is limited use of data-driven approaches to predict worker fatigue or optimize task scheduling within the plantation context.

This research proposes a Labor Management and Welfare Optimization System as a core component of an AI-Driven Intelligent Decision Support System for Sri Lankan tea estate management. The proposed system leverages machine learning techniques to analyze labor-related data, including attendance records, working hours, health indicators, task performance history, and grievance submissions collected through a dedicated web application. Machine learning models are trained to detect workload imbalances and predict worker fatigue levels, enabling timely and proactive interventions by estate managers.

In addition to fatigue prediction, the system incorporates an AI-based task allocation engine that assigns appropriate tasks to workers based on their physical capability, health status, experience level, and availability. Furthermore, a Natural Language Processing (NLP)-based grievance categorization module automatically classifies and prioritizes worker grievances, enabling faster response and improved transparency in grievance management. The outputs of the system include optimized daily task schedules, worker welfare alerts, and prioritized grievance reports, collectively supporting both ethical labor management and operational efficiency.

The proposed solution addresses a significant gap in existing tea estate labor management systems, where worker welfare monitoring, fatigue prediction, and grievance resolution are rarely supported by intelligent automated technologies. By integrating artificial intelligence, machine learning, and natural language processing into a unified labor decision support platform, this research contributes a novel and practical solution specifically tailored to the operational context of the Sri Lankan tea industry.

### Keywords:

Labor Management, Worker Welfare, Fatigue Prediction, Intelligent Task Allocation, Natural Language Processing, Machine Learning, Decision Support Systems, Tea Estate Management

## **Acknowledgement**

I would like to express my sincere gratitude to my supervisor, Mrs. Uthpala Samarakoon, and my co-supervisor, Ms. Suriyaa Kumari, for their valuable guidance, continuous support, and constructive feedback throughout the preparation of this research proposal. Their expertise, encouragement, and insightful suggestions greatly contributed to improving the quality of this research.

I would also like to extend my appreciation to the Sri Lanka Institute of Information Technology (SLIIT) and the Faculty of Computing for providing the academic environment, knowledge, and resources necessary to conduct this research.

My sincere thanks also go to the lecturers and staff members of the Department of Information Technology for their guidance and support during my undergraduate studies.

Finally, I would like to thank my family and friends for their encouragement, patience, and motivation throughout the completion of this proposal.

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## List of Abbreviations

| Abbreviation | Description                                             |
|--------------|---------------------------------------------------------|
| AI           | Artificial Intelligence                                 |
| ML           | Machine Learning                                        |
| NLP          | Natural Language Processing                             |
| CNN          | Convolutional Neural Network                            |
| DSR          | Design Science Research                                 |
| BERT         | Bidirectional Encoder Representations from Transformers |
| TF-IDF       | Term Frequency – Inverse Document Frequency             |
| LSTM         | Long Short-Term Memory                                  |
| API          | Application Programming Interface                       |
| HRMS         | Human Resource Management System                        |
| SDG          | Sustainable Development Goal                            |
| SLIIT        | Sri Lanka Institute of Information Technology           |
| EPF          | Employees' Provident Fund                               |
| VPS          | Virtual Private Server                                  |

# 1. INTRODUCTION

The tea industry is a cornerstone of Sri Lanka's national economy, providing livelihoods to hundreds of thousands of workers across plantation estates. Tea estates employ a large and diverse labor force, including field workers, pruning teams, nursery staff, and supervisory personnel. Managing this workforce effectively and ethically is critical for maintaining both productivity and the well-being of estate workers, who form the backbone of the industry.

In most Sri Lankan tea estates, labor administration is carried out through traditional manual processes. Attendance records are maintained in physical registers, task assignments are determined by supervisors based on informal assessments, and worker grievances are handled through verbal communication or basic paper-based systems. These methods are often inefficient, inconsistent, and prone to human error. Supervisors frequently lack analytical tools to detect workload imbalances, identify fatigued workers, or make equitable task allocation decisions. As a result, some workers may become overburdened while others remain underutilized, leading to reduced productivity and declining worker welfare.

Worker welfare is an increasing concern in the Sri Lankan tea industry. Prolonged physical exertion, repetitive tasks, inadequate rest periods, and limited access to health support contribute to higher levels of fatigue and occupational health risks among estate workers. Despite these challenges, most existing labor management systems in tea estates focus mainly on attendance tracking and wage calculation, offering little support for welfare monitoring, fatigue prediction, or equitable workload distribution.

Recent advances in Artificial Intelligence (AI), Machine Learning (ML), and Natural Language Processing (NLP) have created significant opportunities to improve labor management in agricultural environments. Machine learning models can analyze historical labor data to identify patterns related to fatigue and workload imbalance, while NLP techniques can automate the classification and prioritization of worker grievances. When combined with intelligent task allocation mechanisms, these technologies can provide estate managers with data-driven decision support tools that enhance both productivity and worker welfare.

This research proposes a Labor Management and Welfare Optimization System as a dedicated component within an AI-Driven Intelligent Decision Support System for Sri Lankan tea estate management. The proposed system collects labor-related data through a web-based application and applies machine learning and NLP algorithms to generate actionable insights for estate managers. The system aims to improve task allocation fairness, detect worker fatigue risks, and streamline grievance resolution, thereby contributing to a more efficient and ethical labor management environment in Sri Lankan tea estates.

## 1.1 Background and Context

Sri Lanka's tea industry is one of the oldest and most well-established in the world, with more than 150 years of continuous plantation activity. The country produces approximately 300 million kilograms of tea annually and is globally recognized for the high quality of Ceylon tea. The industry directly employs over 300,000 estate workers, while many additional dependents rely on plantation-based employment for their livelihoods. The effective management of this large workforce is essential not only for operational productivity but also for the social and economic welfare of plantation communities.

Tea estates are characterized by labor-intensive operations distributed across large land areas. Key activities include tea leaf plucking, pruning, weeding, fertilizer application, nursery maintenance, and transportation. Each of these tasks requires different levels of physical effort, skill, and experience. However, in many estates, task assignments are made informally by field supervisors without a structured, data-driven evaluation of individual worker capacity, health status, or current workload. This approach frequently results in unequal task distribution, where some workers experience excessive workloads while others are assigned tasks below their capability.

The lack of structured welfare monitoring mechanisms in tea estates further intensifies these challenges. Worker fatigue is a significant occupational health concern, particularly for field workers engaged in physically demanding tasks. Chronic fatigue not only reduces individual productivity but also increases the risk of work-related injuries and long-term health complications. Despite the seriousness of this issue, most tea estates lack systematic tools for monitoring or predicting worker fatigue levels.

Grievance management represents another critical concern in plantation environments. Tea estate workers frequently encounter issues related to wages, working conditions, task assignments, and interpersonal conflicts. In many estates, grievance reporting relies on informal verbal communication or basic written complaints submitted to supervisors. These unstructured processes often result in delayed responses, inconsistent resolutions, and, in some cases, the suppression of legitimate worker concerns. The absence of automated grievance classification and prioritization mechanisms further limits the ability of managers to identify and address systemic issues.

This research aligns with Sustainable Development Goal (SDG) 8: Decent Work and Economic Growth and SDG 3: Good Health and Well-Being by promoting technology-driven improvements in labor welfare and workforce management within Sri Lankan tea estates. The study also belongs to the Technology Integration and Management (TIM) research cluster, which focuses on applying artificial intelligence technologies to address practical labor management challenges in agricultural environments.

## 1.2 Literature Review

Artificial Intelligence (AI) and Machine Learning (ML) technologies have increasingly been applied in workforce management across multiple industries, demonstrating significant potential for improving labor efficiency, health monitoring, and decision support systems. Recent studies also highlight the growing use of AI technologies in agricultural management and productivity optimization. Kamilaris and Prenafeta-Boldú (2018) conducted a comprehensive survey on the application of deep learning in agriculture and demonstrated how AI techniques can support decision making, crop monitoring, and agricultural productivity improvement [1]. Their findings indicate that AI-driven analytical systems can significantly enhance agricultural management practices by enabling data-driven insights.

Machine learning has also been widely applied to workforce management and productivity analysis. Sarker et al. (2021) proposed a machine learning-based framework for workforce management that analyzes employee performance data and productivity patterns to support efficient workforce planning [2]. Their research demonstrated that machine learning algorithms can assist organizations in improving productivity, workload distribution, and operational decision making. However, their study primarily focused on general workforce management and did not specifically address labor-intensive agricultural environments.

The **International Labor Organization (ILO)** has also emphasized the importance of digital transformation in agriculture and rural employment. According to the ILO (2022), digital technologies can improve working conditions, productivity, and labor management practices in agricultural sectors by enabling data-driven decision support systems [3]. These technologies can help monitor worker conditions, optimize labor allocation, and support fair labor practices in rural employment contexts.

Several studies have also explored the use of machine learning for occupational health monitoring. Islam et al. (2020) investigated the application of machine learning algorithms for predicting worker fatigue in physically demanding environments. Their study evaluated classifiers such as Random Forest and Support Vector Machines using physiological and activity-based data to predict fatigue levels. The results showed that ensemble machine learning models achieved high prediction accuracy and could be integrated into occupational health monitoring systems [4]. However, the research was conducted in industrial manufacturing environments and did not consider agricultural labor conditions.

Shen et al. (2022) proposed a deep learning-based approach for intelligent workforce scheduling and task allocation in large organizations. Their system analyzed employee performance records, skill profiles, and workload histories to generate optimized task assignments. The system improved workload balance and task efficiency compared to manual scheduling methods [5]. However, the system was designed for office-based workforce environments and did not address the challenges of outdoor agricultural labor management.

Natural Language Processing (NLP) has also been applied in workplace management systems. Liang et al. (2023) developed an NLP-based model for automated classification of employee grievances in large enterprises. Their system processed textual complaint data and classified grievances into predefined categories with high accuracy, significantly reducing the time required for manual grievance handling [6]. Although effective, this approach was designed for corporate environments and did not account for the domain-specific language used by plantation workers.

Research focusing specifically on Sri Lankan tea estates has also highlighted significant labor management challenges. Mendis and Fernando (2021) examined labor administration practices in tea plantations and found that most estates rely heavily on manual registers and supervisor-based decision making for attendance tracking and task allocation [7]. This lack of digital tools results in inconsistent task assignments and limited monitoring of worker welfare.

Similarly, Rajapaksha et al. (2022) investigated the relationship between workload distribution and worker productivity in Sri Lankan tea estates. Their findings revealed that workload imbalance significantly contributes to worker fatigue, reduced yield quality, and increased absenteeism. The study recommended the implementation of data-driven labor management tools to improve productivity and worker welfare [8].

In addition, Kumar and Singh (2019) proposed a rule-based intelligent task allocation framework using NLP governance rules for workforce management. Their system allowed managers to express task assignment policies in natural language, which were automatically translated into machine-readable rules for scheduling decisions [9]. While this approach demonstrated strong potential for policy-driven workforce management, its application in agricultural labor environments has not been extensively explored.

In addition to academic research, several commercial platforms have been developed for agricultural workforce management. Tools such as AgriWebb, Croptracker, and general-purpose Human Resource Management Systems (HRMS) provide digital attendance tracking and basic workforce analytics. However, these systems are primarily designed for general agricultural management and do not incorporate advanced features such as AI-based fatigue prediction, NLP-based grievance processing, or intelligent task allocation tailored to tea estate operations.

Overall, the existing literature demonstrates the growing potential of machine learning, NLP, and intelligent decision support technologies in workforce management. However, there remains a significant research gap in developing integrated AI-driven systems specifically designed for tea estate labor management. In particular, the combined use of fatigue prediction, intelligent task allocation, and NLP-based grievance management within the Sri Lankan plantation context has not been adequately explored.

### 1.3 Research Gap

Although existing research has demonstrated the effectiveness of Artificial Intelligence (AI), Machine Learning (ML), and Natural Language Processing (NLP) techniques in workforce management and labor analytics, a significant gap remains in applying these technologies to the specific labor management challenges faced in Sri Lankan tea estates.

Firstly, most existing workforce management systems focus primarily on administrative functions such as attendance tracking and wage calculation. While these systems support basic record management, they do not incorporate predictive analytics for monitoring worker fatigue or intelligent mechanisms for optimizing workload distribution. Tea estate labor involves prolonged outdoor work, physically demanding tasks, and varying environmental conditions. These factors require a more advanced and welfare-oriented labor management approach that integrates predictive analytics and data-driven decision support.

Secondly, although NLP-based grievance management systems have been developed for corporate and enterprise environments, there is limited research addressing grievance reporting mechanisms specifically designed for plantation workers. Tea estate workers may have varying levels of digital literacy, and their grievances often involve plantation-specific terminology and work conditions. Therefore, a grievance management system tailored to the linguistic and operational context of tea estates is required, supported by domain-specific NLP models.

Thirdly, existing AI-based task allocation systems are primarily designed for structured office or industrial environments where employee roles, task requirements, and worker capabilities are relatively standardized. In contrast, tea estate operations involve diverse and dynamic tasks influenced by seasonal variations, field conditions, worker health status, and levels of experience. These factors create a more complex task allocation challenge that existing systems do not adequately address.

Therefore, there is a clear need for an integrated intelligent labor management system specifically designed for tea estate environments. Such a system should combine machine learning-based fatigue prediction, AI-driven task allocation, and NLP-based grievance categorization within a unified decision support platform to support both workforce productivity and worker welfare in Sri Lankan tea estates.

**Table 1.1** presents a comparison between existing labor management systems and the proposed AI-based solution. It highlights the limitations of current systems, which mainly focus on attendance and wage tracking, while the proposed method integrates advanced features such as fatigue prediction, intelligent task allocation, grievance analysis, and decision support tailored for plantation environments.

*Table 1.1 Comparison of existing research methods and the proposed method*

| <b>Technologies and Methods</b>    | <b>Existing System 1: Attendance &amp; Wage Systems</b> | <b>Existing System 2: ERP Labor Modules</b> | <b>Existing System 3: General Workforce Applications</b> | <b>Proposed Method</b> |
|------------------------------------|---------------------------------------------------------|---------------------------------------------|----------------------------------------------------------|------------------------|
| Attendance and Wage Tracking       | ✓                                                       | ✓                                           | ✓                                                        | ✓                      |
| Worker Health Monitoring           | ✗                                                       | ✗                                           | ✗                                                        | ✓                      |
| AI-Based Fatigue Prediction        | ✗                                                       | ✗                                           | ✗                                                        | ✓                      |
| Intelligent Task Allocation        | ✗                                                       | ✗                                           | ✗                                                        | ✓                      |
| NLP-Based Grievance Categorization | ✗                                                       | ✗                                           | ✗                                                        | ✓                      |
| Worker Welfare Analytics           | ✗                                                       | ✓                                           | ✗                                                        | ✓                      |
| Plantation-Specific Adaptation     | ✗                                                       | ✗                                           | ✗                                                        | ✓                      |
| Integrated Decision Support        | ✗                                                       | ✓                                           | ✗                                                        | ✓                      |

## **2. OBJECTIVE**

### **2.1 Main Objective**

To design and develop an intelligent Labor Management and Welfare Optimization System that applies machine learning and natural language processing techniques to predict worker fatigue, optimize task allocation, and automate grievance categorization in Sri Lankan tea estates.

### **2.2 Specific Objectives**

1. To collect and analyze labor-related data from tea estates, including attendance records, working hours, health indicators, task performance history, and grievance submissions, in order to identify patterns of workload imbalance and potential fatigue risk factors.
2. To design and train machine learning models capable of predicting worker fatigue levels using collected labor data, enabling estate managers to implement proactive welfare interventions.
3. To develop an AI-based task allocation recommendation engine that assigns appropriate tasks to workers based on factors such as health status, physical capability, experience level, and workload history.
4. To implement a Natural Language Processing (NLP)-based grievance categorization and prioritization system that automatically classifies worker grievance submissions and assigns priority levels to improve response efficiency.
5. To integrate all system components into a web-based estate manager dashboard that provides real-time visibility of workforce conditions, task allocations, fatigue alerts, and grievance management status.
6. To evaluate the performance of the proposed system using appropriate metrics and validate its effectiveness in improving labor management efficiency, workload balance, and worker welfare in tea estates.

## **3. METHODOLOGY**

### **3.1 Research Approach**

This research adopts the Design Science Research (DSR) methodology, which focuses on designing, developing, and evaluating technological artifacts that address real-world problems. In this study, the proposed artifact is the Labor Management and Welfare Optimization System, which provides intelligent decision support for estate managers using machine learning and Natural Language Processing (NLP) techniques.

The research process consists of five main phases: problem identification and analysis, system design and architecture planning, data collection and model development, system implementation and integration, and system evaluation and validation. The iterative nature of the DSR approach allows continuous improvement of the system based on testing results and feedback.

To ensure practical relevance, field research activities will involve interactions with tea estate managers, field supervisors, and workers. These interactions will help collect primary data and domain knowledge, ensuring that the proposed system is designed according to the real operational conditions of Sri Lankan tea estates.

### **3.2 Technical Methods and Algorithms**

The proposed Labor Management and Welfare Optimization System employs three main technical approaches: machine learning for fatigue prediction, AI-based task allocation, and Natural Language Processing (NLP) for grievance categorization. These components work together to provide intelligent decision support for tea estate labor management.

#### **Fatigue Prediction Model**

Worker fatigue prediction is implemented using supervised machine learning algorithms. Algorithms such as Random Forest and Gradient Boosting are evaluated for their effectiveness in predicting fatigue levels. The models use input features including cumulative working hours, attendance frequency, task intensity scores, health indicators, and historical fatigue records.

Training data is collected from estate labor records and annotated with fatigue indicators identified by domain experts. The trained model analyzes patterns in worker activity and predicts fatigue risk levels, allowing estate managers to take preventive actions and adjust workloads when necessary.

#### **Task Allocation Engine**

The intelligent task allocation engine combines rule-based optimization with NLP-parsed governance rules. Estate managers can define task assignment policies using natural language instructions. For example, experienced workers may be assigned complex tasks such as pruning, while workers with health restrictions may be limited to lighter duties.

The NLP module interprets these natural language policies and converts them into machine-readable rules. These rules are then applied together with worker profile data, fatigue scores, and availability information to generate optimized daily task allocation recommendations.

### **NLP-Based Grievance Categorization**

The grievance categorization module processes worker grievance submissions using Natural Language Processing techniques. Text preprocessing steps include tokenization, stop-word removal, and stemming to prepare the text for analysis.

A text classification model, using techniques such as TF-IDF feature extraction with Logistic Regression or a fine-tuned BERT model, is used to categorize grievances into predefined classes. These categories may include wage issues, working conditions, health and safety concerns, task allocation disputes, and interpersonal conflicts.

Each grievance is automatically assigned a priority level based on severity keywords and predefined category weights. This enables estate managers to quickly identify critical issues and respond more efficiently.

### **3.3 Tools and Platforms**

The system is developed using technologies that support machine learning, Natural Language Processing (NLP), and web application development. Python is used for machine learning and NLP implementation, with libraries such as scikit-learn, TensorFlow, NLTK, and spaCy.

The web application is built using React for the frontend and Node.js for backend services and API development. A database such as MongoDB or PostgreSQL is used to store worker data, task records, and grievance information.

The estate manager dashboard is designed as a responsive web interface accessible from desktop and mobile devices. The system can be deployed on cloud platforms such as Google Cloud or AWS to support scalable processing and secure data storage.

### **3.4 Data Collection and Ethical Considerations**

Primary data will be collected from selected tea estates through field visits, supervisor interviews, and reviews of existing labor records. The collected data will include worker attendance records, working hours, task allocation details, health records maintained by estate dispensaries, and grievance submissions from workers.

Secondary data will be obtained from published reports and studies from organizations such as the Employees' Provident Fund (EPF), the Tea Research Institute of Sri Lanka, and other academic sources to support analysis of labor patterns and welfare conditions in the tea industry.

All data collection activities will follow ethical research guidelines. Worker personal information will be anonymized before analysis, and the data will be used only for research and model development purposes. Formal permission will be obtained from participating estates, and informed consent will be collected from all participants in accordance with the ethical standards of Sri Lanka Institute of Information Technology (SLIIT).

### **3.5 Validation Strategy and Performance Metrics**

The performance of the proposed machine learning and NLP models will be evaluated using appropriate evaluation metrics.

The fatigue prediction model will be assessed using accuracy, precision, recall, F1-score, and AUC-ROC. The dataset will be divided into training, validation, and testing sets, and cross-validation will be applied to ensure reliable model performance.

The NLP grievance categorization model will be evaluated using classification accuracy, precision, recall, F1-score, and confusion matrix analysis to measure classification performance across grievance categories.

The task allocation engine will be evaluated through simulation studies that compare automated task allocation results with traditional manual assignment methods. In addition, user acceptance testing with estate managers will be conducted to assess system usability and practical effectiveness.

## **4. HIGH-LEVEL SYSTEM ARCHITECTURE**

### **4.1 Overview of the Proposed System**

The proposed system represents the Labor Management and Welfare Optimization component of an AI-driven Intelligent Decision Support System for Sri Lankan tea estate management. The overall architecture consists of four main components: Labor Management and Welfare Optimization, Tea Leaf Disease Detection and Pattern Analysis, Tea-Specific Climate and Weather Intelligence, and AI-based Tea Sorting and Grading.

The Labor Management and Welfare Optimization component functions as the workforce management subsystem, responsible for labor data processing, predictive analytics, and decision support. Worker data is collected through a web-based interface used by estate managers and supervisors.

This data is processed by machine learning and NLP modules on the backend to generate outputs such as task allocation recommendations, fatigue alerts, and categorized grievance reports, which are displayed through the estate manager dashboard.

### **4.2 System Components**

#### **1. Data Collection Layer**

A web-based application used by estate managers and supervisors to record labor data such as attendance, task assignments, health updates, and grievance submissions. The interface is designed to function efficiently in low-connectivity plantation environments.

#### **2. Worker Profile Management Module**

Maintains a database of worker profiles including demographic information, health indicators, experience, training records, attendance history, and performance data. This module provides input data for the fatigue prediction and task allocation modules.

#### **3. Machine Learning Fatigue Prediction Module**

Analyzes historical and real-time labor data to estimate worker fatigue risk levels. The system generates alerts when workers show high fatigue risk, allowing supervisors to adjust workloads.

#### **4. AI-Based Task Allocation Module**

Generates optimized task assignments by combining worker profile data, fatigue scores, and governance rules interpreted using NLP. This ensures balanced workload distribution while considering worker capability and health conditions.

## **5. NLP Grievance Categorization Module**

Processes worker grievance submissions using Natural Language Processing techniques and classifies them into predefined categories with priority levels for faster resolution.

## **6.Estate Manager Dashboard**

A web-based interface that displays workforce insights such as task schedules, fatigue alerts, grievance reports, and productivity analytics. Managers can review and adjust system recommendations when necessary.

## **7. Database and Storage Layer**

A relational or NoSQL database stores worker profiles, labor records, grievance data, system outputs, and logs, supporting both real-time operations and historical analysis.

### **4.3 Component Interaction**

The system components communicate through a structured data flow managed by the backend API. Estate managers and supervisors update worker records through the web interface, and this data is stored in the worker profile database.

The fatigue prediction module analyzes updated labor data and generates fatigue risk scores for workers. These scores are sent to the dashboard alert system and the task allocation module, which uses worker profiles, fatigue scores, and NLP-based governance rules to recommend appropriate task assignments.

Worker grievances submitted through the system are processed by the NLP categorization module, which classifies and prioritizes each complaint before displaying it in the grievance management section of the manager dashboard.

All system outputs are stored in the database to support historical analysis, system monitoring, and future model improvements

### **4.4 Data Flow Explanation**

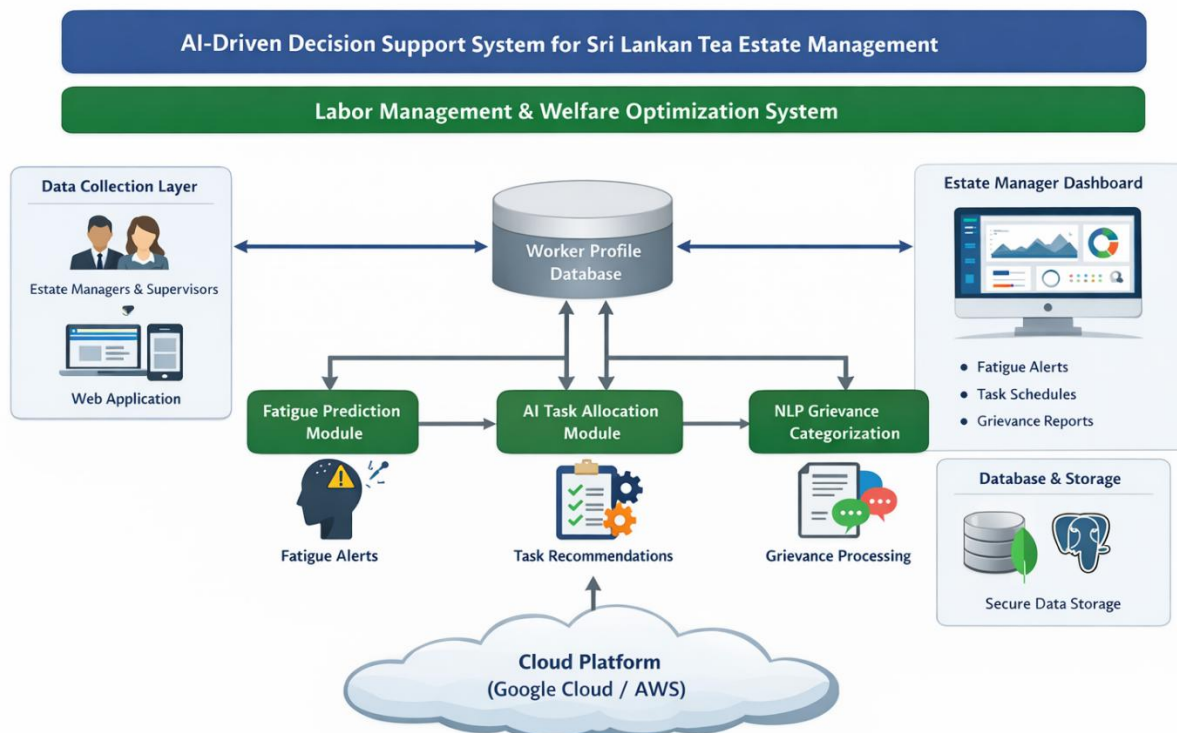
1. Estate managers or supervisors enter worker attendance, health status, and task information through the web application.
2. The worker profile module updates and stores worker records in the database.
3. The fatigue prediction model analyzes updated labor data and generates fatigue risk scores for each worker.
4. Workers with high fatigue risk are flagged, and welfare alerts are sent to the manager dashboard.

5. The task allocation engine uses worker profiles, fatigue scores, and governance rules to generate recommended daily task assignments.
6. Task schedules are displayed on the manager dashboard, where managers can review and adjust them if necessary.
7. Worker grievances submitted through the system are processed by the NLP categorization module.
8. Classified grievances and priority levels are displayed in the grievance management panel of the dashboard.
9. All system outputs are stored in the database for historical analysis, reporting, and model performance monitoring.

#### 4.5 Scalability and Security Considerations

The system architecture is designed to support scalability and flexibility. Its modular structure allows components such as the fatigue prediction model and NLP engine to be updated or expanded without affecting other system modules. Cloud-based deployment enables the system to scale efficiently as more tea estates and worker data are added.

Security is ensured through multiple measures. Role-based access control limits dashboard access to authorized users. Worker data is stored securely with data encryption, and communication



*Figure 4.1 System Architecture Diagram*

between the web interface and backend server is protected using HTTPS protocols. Additionally, worker information is anonymized to ensure compliance with data privacy regulations and SLIIT ethical guidelines.

## **5. USER REQUIREMENTS**

### **5.1 Stakeholder Identification**

The proposed **Labor Management and Welfare Optimization System** involves several stakeholders who interact with the system or benefit from its outputs.

#### **Estate Managers**

Estate managers are the primary users of the system. They use the manager dashboard to monitor workforce conditions, review task schedules, receive fatigue alerts, and manage worker grievances.

#### **Field Supervisors**

Field supervisors work directly with estate workers and use the system to access task allocations, monitor worker conditions, and update attendance and health information.

#### **Tea Estate Workers**

Workers are indirect beneficiaries of the system. They benefit from fair task assignments, improved welfare monitoring, and a structured grievance reporting process.

#### **Estate Dispensary and Health Staff**

Health staff provide worker health information used for fatigue prediction and receive alerts when workers may require medical attention or reduced workloads.

#### **Research Team and System Developers**

Researchers analyze labor data, train machine learning models, and evaluate system performance, while developers maintain and improve the system infrastructure.

### **5.2 Functional Requirements**

FR1: The system shall allow estate managers and supervisors to create and update worker profiles, including demographic details, health status, experience, and task history.

FR2: The system shall collect and store daily labor data such as attendance, working hours, task assignments, and health indicators.

FR3: The system shall analyze worker data using machine learning models to predict fatigue risk levels and generate welfare alerts.

FR4: The system shall generate optimized task allocation recommendations based on worker profiles, fatigue scores, and governance rules.

FR5: The system shall allow workers to submit grievances and automatically categorize them using NLP techniques.

FR6: The system shall provide a manager dashboard that displays task schedules, fatigue alerts, grievance reports, and workforce analytics.

### **5.3 Non-Functional Requirements**

#### **Security**

The system shall implement role-based access control to protect sensitive worker data. All communications must use HTTPS encryption, and personal information must be securely stored to prevent unauthorized access.

#### **Performance**

The fatigue prediction model shall generate updated risk scores quickly after data input, and the task allocation engine shall produce daily task recommendations within a short processing time.

#### **Usability**

The manager dashboard and supervisor interface shall be simple and intuitive, requiring minimal training. The worker grievance interface shall support users with limited digital literacy through clear forms and instructions.

#### **Reliability**

The system shall maintain stable performance and accurate predictions across different data volumes and worker populations, ensuring continuous availability during estate working hours.

#### **Scalability**

The system shall support the addition of new tea estates, workers, and data without significant performance degradation.

#### **Maintainability**

The system shall use a modular architecture so that machine learning models and NLP components can be updated or improved independently.

## **6. COMMERCIALIZATION PLAN**

### **6.1 Target Market and Customer Persona**

The primary target market for the proposed Labor Management and Welfare Optimization System includes plantation companies, regional tea estate management organizations, and large private tea estates in Sri Lanka. These organizations manage large numbers of workers and require efficient tools to monitor workforce conditions, improve productivity, and ensure worker welfare.

The main customer persona includes estate managers and plantation supervisors responsible for workforce planning, task allocation, and labor welfare monitoring. These users require reliable and easy-to-use digital tools that support data-driven decision making, worker fatigue monitoring, and efficient grievance management.

Secondary users include government labor authorities, the Sri Lanka Tea Board, labor welfare organizations, and academic researchers, who may use aggregated data to study labor patterns, worker welfare conditions, and productivity trends within the plantation sector.

### **6.2 Value Proposition**

The proposed system provides an AI-driven labor management solution designed specifically for tea estate environments. Unlike traditional manual labor administration methods, the system enables data-driven workforce management and welfare monitoring.

Key value propositions include:

- AI-based fatigue prediction to identify workers at risk of exhaustion and prevent health issues
- Intelligent task allocation to ensure fair workload distribution among workers
- NLP-based grievance categorization to improve the efficiency of worker complaint handling
- Real-time workforce monitoring dashboard for estate managers
- Data-driven labor analytics to support better management decisions

### **6.3 Revenue Model**

The proposed system can adopt a Software-as-a-Service (SaaS) model where tea estates subscribe to the platform on a monthly or annual basis.

Possible revenue streams include:

- Subscription plans for plantation companies and tea estates
- Premium workforce analytics and reporting features for large estates
- Custom system integration services for plantation management systems
- Data analytics services for research institutions and government agencies

## 6.4 Cost Estimation

The main costs involved in developing and deploying the system include:

- System Development Costs – machine learning model development, software engineering, and testing
- Infrastructure Costs – cloud hosting, database management, and server processing
- Data Collection Costs – gathering and processing labor records and workforce datasets
- Maintenance Costs – system updates, model retraining, and technical support

## 6.5 Pricing Strategy

A tiered pricing strategy can be implemented based on the size of the estate and system usage. Example pricing structure:

- **Basic Plan** – Small tea estates with core workforce monitoring and grievance management features
- **Standard Plan** – Medium estates with AI fatigue prediction, task optimization, and reporting tools
- **Enterprise Plan** – Large plantation companies with advanced analytics, unlimited worker records, and customized dashboards

## 6.6 Competitive Advantage

The proposed system offers several advantages compared to existing labor management tools:

- Designed specifically for tea estate workforce management
- Integration of fatigue prediction, task allocation, and grievance management in one platform
- AI-based decision support for workforce planning and welfare monitoring
- More cost-effective compared to complex enterprise HR management systems

## 6.7 Intellectual Property Considerations

The system may include several intellectual property components, including:

- Proprietary machine learning models for worker fatigue prediction
- Custom NLP models for grievance classification
- The integrated AI-based labor decision support platform architecture
- Governance rule framework for AI-assisted task allocation

## 7. BUDGET AND JUSTIFICATION

The following budget covers all hardware, software, cloud services, and human effort costs required for the development, deployment, and validation of the Labor Management and Welfare Optimization System.

*Table 7.1 Estimated budget table*

| <b>Description</b>                                                         | <b>Total Amount (Rs.)</b> |
|----------------------------------------------------------------------------|---------------------------|
| Hardware – Laptop/workstation upgrade for model training and development   | 8,000                     |
| Cloud Hosting for ML Model Training and Deployment (6 months – AWS/GCP)    | 10,500                    |
| Software Licenses – Python ML libraries, NLP frameworks, IDE subscriptions | 3,000                     |
| Web Application Hosting (Node.js Backend Server – VPS)                     | 6,000                     |
| Database Hosting (MongoDB / PostgreSQL Cloud Service)                      | 3,500                     |
| Human Effort – Development and testing labor (estimated)                   | 4,000                     |
| Internet & Utilities for Development (shared component cost)               | 2,000                     |
| Transport for Field Data Collection (tea estate visits)                    | 5,500                     |
| Survey Instruments and Questionnaire Preparation                           | 1,500                     |
| Miscellaneous (storage devices, adapters, and unexpected costs)            | 2,000                     |
| <b>Total Budget</b>                                                        | <b>46,000</b>             |

## 8. WORK BREAKDOWN STRUCTURE (WBS)

Table 8.1 presents the project task schedule for the proposed system. It outlines the key activities, timeline, and expected deliverables from literature review and data collection to model development, system integration, testing, and final documentation, providing a structured plan to guide the development and implementation of the AI-based labor management system.

*Table 8.1 Work Breakdown Structure*

| <b>Task ID</b> | <b>Task Description</b>                                                            | <b>Timeline</b> | <b>Deliverable</b>          |
|----------------|------------------------------------------------------------------------------------|-----------------|-----------------------------|
| T1             | Conduct literature review on AI-based labor management and NLP systems             | March           | Literature review report    |
| T2             | Collect labor data from tea estates including attendance, health, and task records | March – April   | Labor dataset               |
| T3             | Analyze collected labor data and identify fatigue patterns                         | April           | Data analysis report        |
| T4             | Design and train machine learning model for fatigue prediction                     | May             | Trained ML model            |
| T5             | Develop AI-based task allocation module                                            | May – June      | Task allocation engine      |
| T6             | Implement NLP-based grievance categorization system                                | June – July     | NLP classification module   |
| T7             | Evaluate system performance using ML evaluation metrics                            | July            | Evaluation report           |
| T8             | Design and develop estate manager web dashboard                                    | August          | Dashboard prototype         |
| T9             | Integrate ML, NLP, and dashboard components into full system                       | September       | Integrated system prototype |
| T10            | Conduct system testing and validation with estate users                            | October         | Testing report              |
| T11            | Prepare final research report and documentation                                    | November        | Final dissertation          |

The Work Breakdown Structure (WBS) illustrates the hierarchical decomposition of the **AI-Based Labor Management System for Tea Plantations** project. It organizes the project into major components including research, model development, system integration, and evaluation. Each component is further divided into sub-modules and specific tasks (T1–T11) to clearly define the development process.

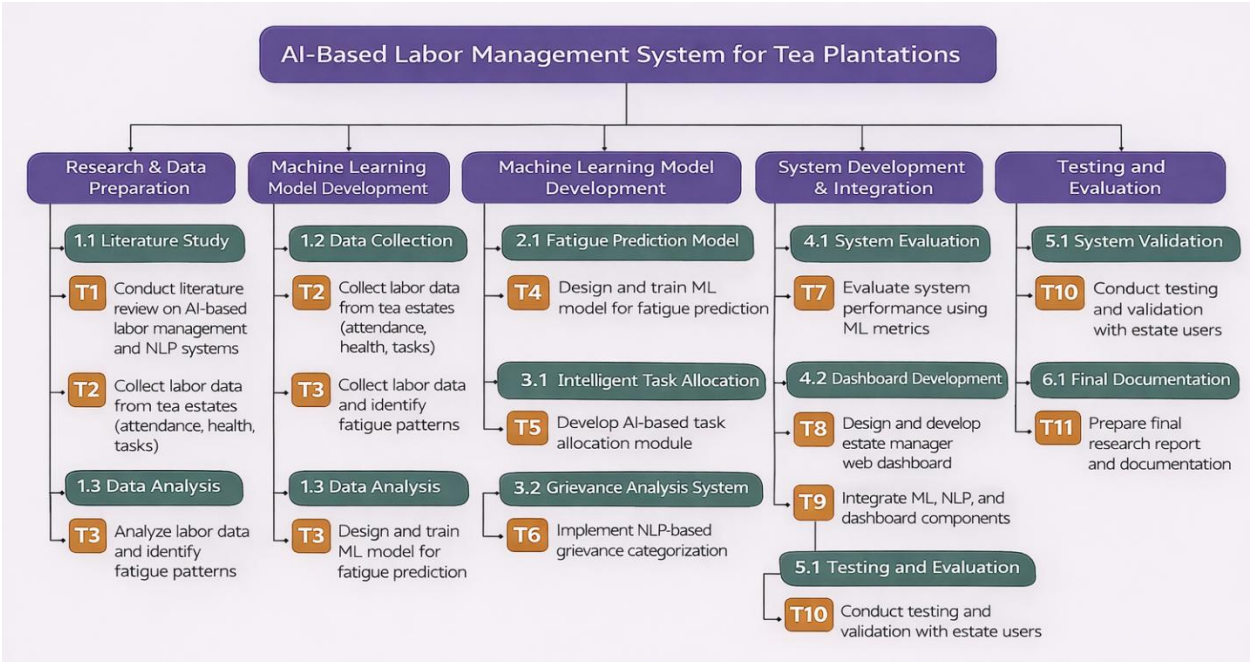
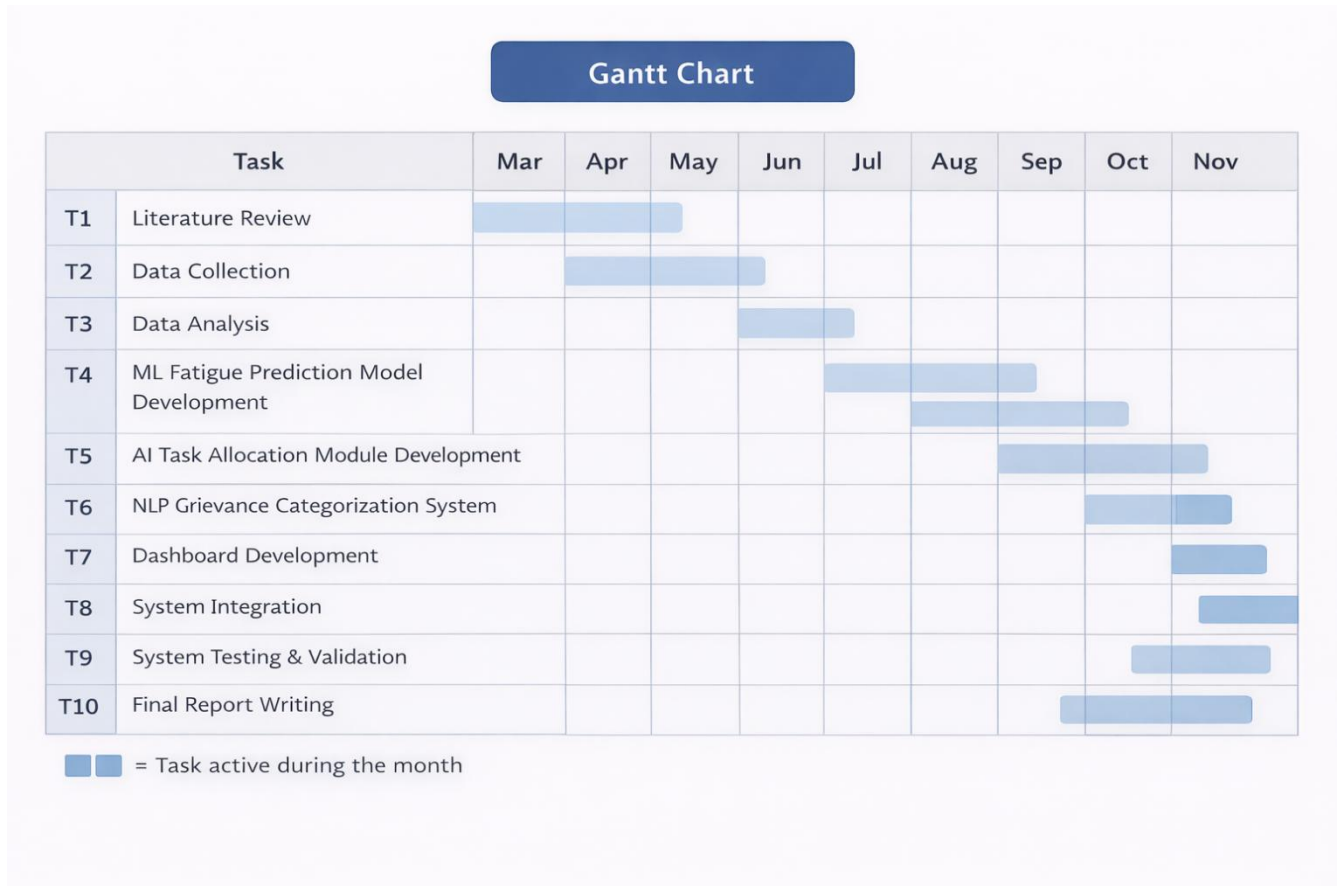


Figure 8.1 Work Breakdown Structure

## 9. GANTT CHART

The project timeline is illustrated in Figure 9.1, which presents the Gantt chart showing the planned schedule for all research tasks across the project period from March to November 2026.



*Figure 9.1 Project Gantt Chart*

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## **11. APPENDICES**

### **11.1 Survey Instrument**

#### **Section A – Respondent Information**

1. What is your role in the tea estate?
  - Estate Manager
  - Field Supervisor
  - Plantation Worker
  - Estate Dispensary / Health Staff
2. How many years of experience do you have in tea plantation management?
  - Less than 5 years
  - – 10 years
  - More than 10 years

#### **Section B – Current Labor Management Practices**

3. How are daily task assignments currently determined in your estate?
  - Supervisor decision based on experience
  - Formal written task schedules
  - Rotational system
  - Other: \_\_\_\_\_
4. Are worker fatigue or health conditions considered when assigning tasks?
  - Always
  - Sometimes
  - Rarely
  - Never
5. How are worker grievances currently collected and processed?
  - Verbal submission to supervisor
  - Written complaint form
  - Through union representatives
  - Other: \_\_\_\_\_

#### **Section C – Challenges in Labor Management**

6. What are the main challenges in managing worker welfare and task assignments?
  - Unequal workload distribution among workers
  - Difficulty identifying fatigued or unwell workers
  - Delayed response to worker grievances
  - Lack of data-driven tools for labor management
  - Insufficient record-keeping systems

7. How often do you observe workload imbalance among estate workers?

- Frequently
- Sometimes
- Rarely

### Section D – Technology Adoption

8. Would an AI-based system that predicts worker fatigue and recommends task assignments be useful for your estate?

- Yes
- No
- Not sure

9. Would you be willing to use a web-based dashboard for labor management and welfare monitoring?

- Yes
- No

10. Which features would you prioritize in a digital labor management system?  
Fatigue prediction and welfare alerts

- Automated task allocation recommendations
- Grievance submission and tracking system
- Attendance and productivity analytics
- Worker health monitoring dashboard

## 11.2 Detailed Diagrams

This section provides detailed process diagrams for the three core intelligent modules of the proposed system. Figure 11.2.1 illustrates the machine learning fatigue prediction pipeline, showing the data flow from input features through model processing to fatigue risk output.

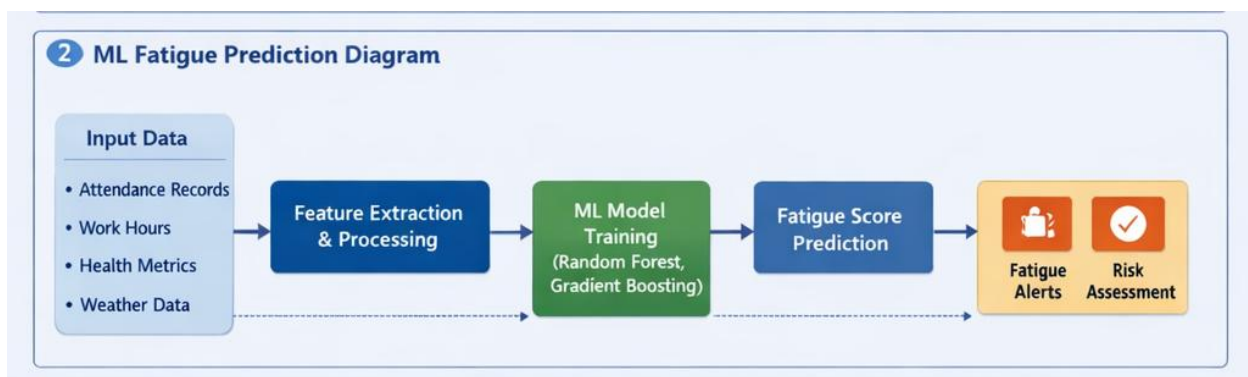


Figure 11.2.1 ML Fatigue Prediction

As shown in Figure 11.2.1, the fatigue prediction process begins with input data collection, passes through feature extraction and model inference, and produces a fatigue risk score used to generate welfare alerts for supervisors. Figure 11.2.2 illustrates the NLP pipeline used for automated grievance categorization.

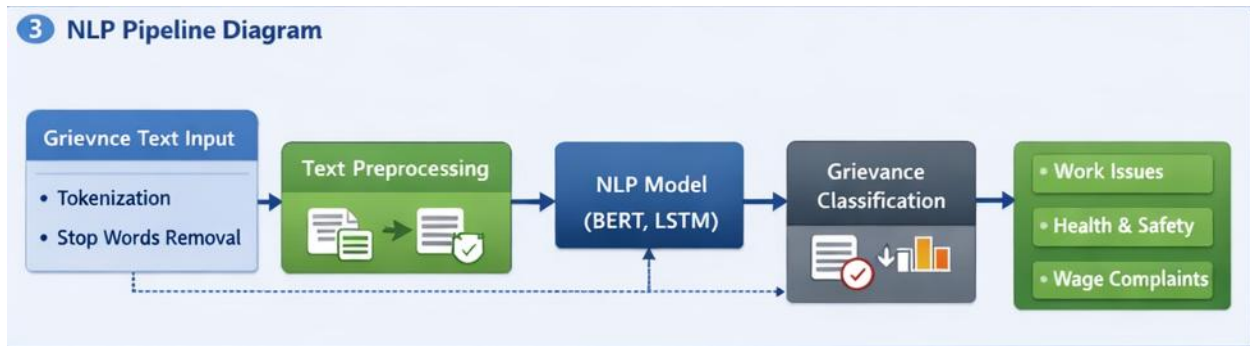


Figure 11.2.2 NLP Pipeline

As shown in Figure 11.2.2, the NLP pipeline processes raw grievance text through tokenization, stop-word removal, and feature extraction before classifying each submission into a grievance category and assigning a priority level. Figure 11.2.3 presents the task allocation flow diagram showing how worker profiles, fatigue scores, and governance rules are combined to generate daily task recommendations.

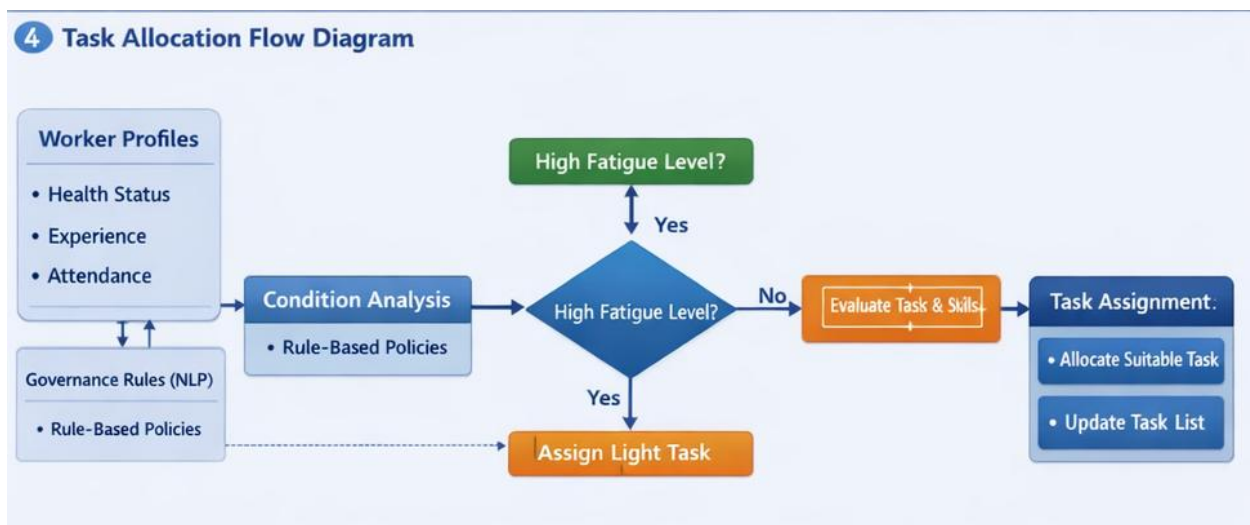


Figure 11.2.3 Task Allocation Flow Diagrams

As illustrated in Figure 11.2.3, the task allocation engine evaluates worker availability, health status, fatigue risk, and experience level against available tasks, and applies NLP-parsed governance rules to generate the final recommended daily task schedule.

### 11.3 Preliminary UI Mockups

The following UI mockups illustrate the key interface screens of the Labor Management and Welfare Optimization web application. These preliminary designs demonstrate the layout and functionality of the estate manager dashboard, worker profile management interface, task allocation view, grievance management panel, and welfare alert notifications.

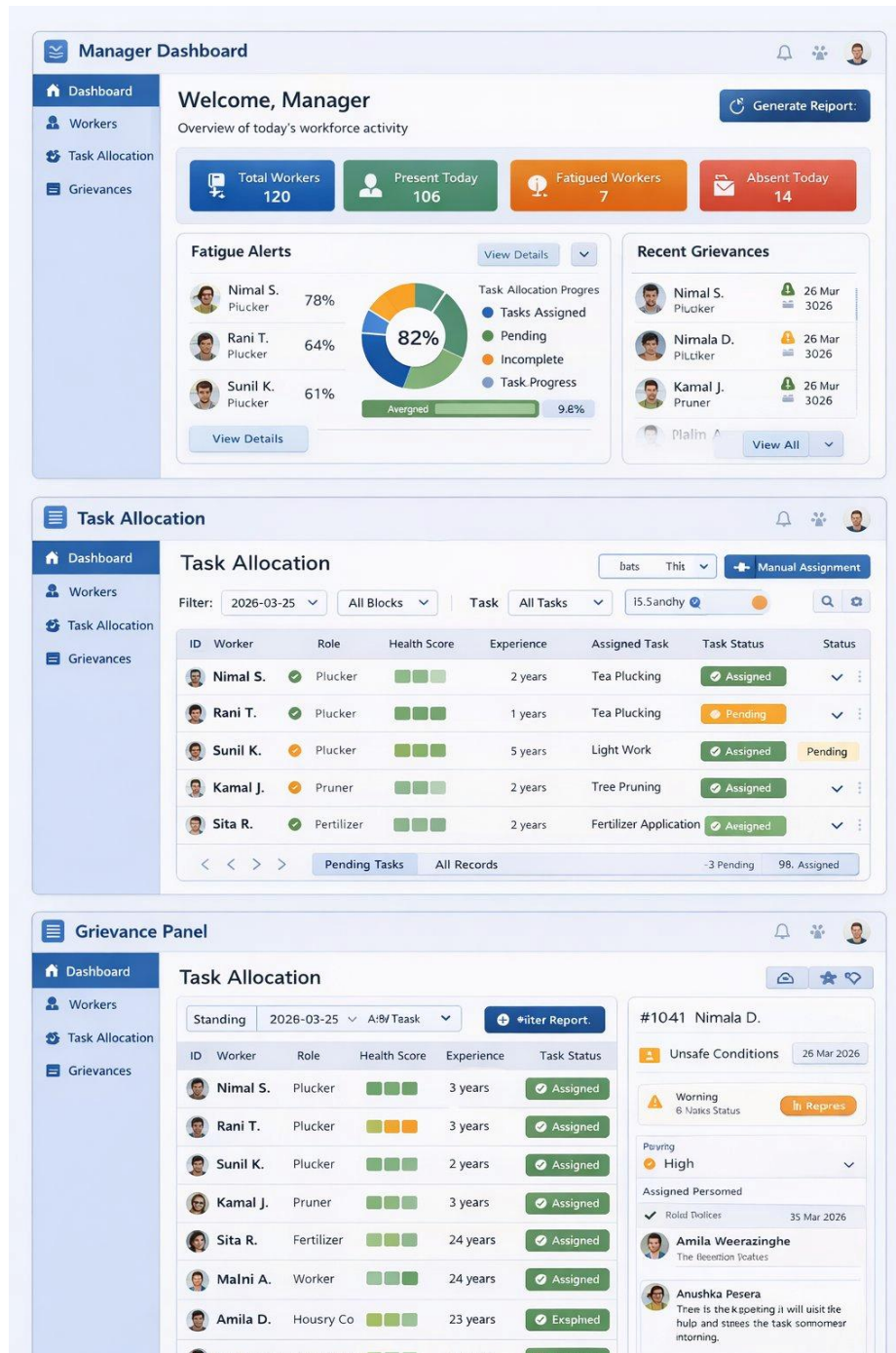


Figure 11.3.1 – Manager Dashboard, Task Allocation Interface & Grievance Panel UI Mockups

## 11.4 Risk Analysis

This section identifies the key risks associated with the proposed research and outlines appropriate mitigation strategies for each. Table 11.1 summarizes the identified risks, their impact levels, and corresponding mitigation approaches.

*Table 11.1 Risk Analysis Table*

| <b>Risk ID</b> | <b>Risk Description</b>                                           | <b>Impact</b> | <b>Mitigation Strategy</b>                                                                                                |
|----------------|-------------------------------------------------------------------|---------------|---------------------------------------------------------------------------------------------------------------------------|
| R1             | Insufficient or incomplete labor data collected from tea estates  | High          | Collaborate with multiple tea estates and supplement data using historical estate registers and available digital records |
| R2             | Low-quality or inconsistent worker grievance records              | Medium        | Standardize grievance data collection templates and perform data cleaning and preprocessing before model training         |
| R3             | Overfitting in the machine learning fatigue prediction model      | High          | Apply cross-validation techniques, regularization methods, and proper training-validation dataset splitting               |
| R4             | Limited cooperation from estate management or supervisors         | Medium        | Establish formal research agreements and communicate the benefits of the system to estate management                      |
| R5             | NLP model inaccuracies when processing domain-specific grievances | Medium        | Fine-tune the NLP model using plantation-specific vocabulary and manually annotated grievance datasets                    |
| R6             | Data privacy concerns related to worker personal information      | High          | Anonymize sensitive worker data and follow SLIIT research ethics guidelines with informed consent                         |

As presented in Table 11.1, the six identified risks range from data collection challenges and model overfitting to privacy concerns and limited estate cooperation. Each risk is assigned a mitigation strategy to ensure the research can proceed effectively and ethically within the project constraints.

## **11.5 Ethical Documents**

The proposed research titled “Labor Management and Welfare Optimization System using Machine Learning and NLP” will follow ethical research standards during all stages of data collection, system development, and evaluation.

### **Informed Consent**

Labor data collected from tea estates will be obtained with the permission of estate management and relevant supervisors. Workers participating in the study will be informed about the purpose of the research and how their data will be used. Written consent forms will be provided in Sinhala and English to ensure that all participants clearly understand the research process.

### **Data Privacy and Confidentiality**

Personal worker information will be handled with strict confidentiality. Identifiable information such as worker names or personal details will be anonymized before processing. Access to collected data will be restricted only to authorized members of the research team.

### **Responsible Use of Data**

All collected data will be used solely for academic research purposes, including training and evaluating machine learning and NLP models. The data will not be shared with external organizations or used for commercial purposes without proper consent.

### **Worker Welfare Protection**

The research is designed to support and improve worker welfare within tea estates. System outputs such as fatigue alerts and task allocation recommendations are intended to promote fair workload distribution and worker well-being rather than monitoring or penalizing workers.

### **Data Security**

All research data will be stored in secure and encrypted digital storage systems. Physical documents collected during field visits will be safely stored and destroyed after digitization and verification.

### **Ethical Research Compliance**

This research will follow the ethical guidelines of the Sri Lanka Institute of Information Technology (SLIIT). Necessary ethical approvals will be obtained before conducting field data collection. The study aims to promote ethical labor management practices while protecting the rights and privacy of participating workers.